

Распределение Парето.

$$p(x) = \begin{cases} \frac{\theta-1}{x^\theta}, & x \geq 1 \\ 0, & x < 1 \end{cases} \quad \theta > 1.$$

а) Метод макс. правдоподобия

$$L(\theta) = \prod_{i=1}^n p(x_i, \theta) = (\theta-1)^n \cdot \prod_{i=1}^n x_i^{-\theta}$$

$$\ln L(\theta) = n \ln(\theta-1) - \theta \sum_{i=1}^n \ln(x_i)$$

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \frac{n}{\theta-1} - \sum_{i=1}^n \ln(x_i) = 0 \quad (\text{прим. произв. и нуля})$$

$$\frac{n}{\theta-1} = \sum \ln x_i \Rightarrow \theta-1 = \frac{n}{\sum \ln x_i}$$

$$\bar{\theta} = 1 + \frac{1}{\frac{1}{n} \sum \ln x_i} = 1 + \frac{n}{\sum \ln x_i}$$

$$\frac{\partial^2 \ln L(\theta)}{\partial \theta^2} = -\frac{n}{(\theta-1)^2} < 0 \quad \forall \theta > 1 \Rightarrow \text{лок. макс}$$

Проверка на сильную регулярность (для нормальности ОМП)

$$1) p(x, \theta) = \frac{\theta-1}{x^\theta}, \quad x \geq 1, \quad \theta \in (1, \infty) - \text{открытое.}$$

$$2) p(x, \theta) - \text{непр. дифф. по } \theta \text{ на всей откр. ин. ве.}$$

$$3) \int_{-\infty}^{\infty} \frac{\partial p(x, \theta)}{\partial \theta} dx = \int_1^{\infty} \frac{\ln x - \theta \ln x}{x^\theta} dx = \frac{x^{1-\theta}}{1-\theta} \Big|_1^{\infty}$$

$$\int_1^{\infty} \ln x d(x^{1-\theta}) = \frac{1}{\theta-1} - \frac{1}{\theta-1} = 0$$

$$4) I(\theta) = M \left[\frac{\partial \ln p(x, \theta)}{\partial \theta} \right]^2$$

$$\ln p(x, \theta) = \ln((\theta-1) \cdot x^{-\theta}) = \ln(\theta-1) - \theta \ln x$$

$$\frac{\partial \ln p(x, \theta)}{\partial \theta} = \frac{\partial}{\partial \theta} (\ln(\theta-1)) - \frac{\partial}{\partial \theta} (\theta \ln x) = \frac{1}{\theta-1} - \ln x$$

$$\left(\frac{\partial^2 \ln p(x, \theta)}{\partial \theta^2} = \frac{\partial}{\partial \theta} \left[\frac{1}{\theta-1} - \ln x \right] = -\frac{1}{(\theta-1)^2} \right)$$

$$\left(\frac{1}{\theta-1} - \ln x\right)^2 = \frac{1}{(\theta-1)^2} - \frac{2 \ln x}{\theta-1} + \frac{(\ln x)^2}{1}$$

$$I(\theta) = \int_1^{\infty} \left[\frac{1}{(\theta-1)^2} - \frac{2 \ln x}{\theta-1} + (\ln x)^2 \right] \cdot \frac{\theta-1}{x^\theta} dx$$

$$(t = \ln x, x = e^t, dx = e^t dt)$$

$$I_1 = \int_1^{\infty} \frac{1}{(\theta-1)^2} \cdot \frac{\theta-1}{x^\theta} dx = \frac{1}{\theta-1} \int_1^{\infty} \frac{1}{x^\theta} dx = \frac{1}{\theta-1} \cdot \frac{1}{\theta-1} = \frac{1}{(\theta-1)^2}$$

$$I_2 = - \int_0^{\infty} \frac{2t}{\theta-1} \cdot (\theta-1) \cdot e^{-t(\theta-1)} dt = -2 \int_0^{\infty} t \cdot e^{-t(\theta-1)} dt$$

$$I_2 = -2 \cdot \frac{1}{(\theta-1)^2} = -\frac{2}{(\theta-1)^2}$$

$$I_3 = \int_0^{\infty} t^2 \cdot (\theta-1) e^{-t(\theta-1)} dt = (\theta-1) \int_0^{\infty} t^2 e^{-t(\theta-1)} dt = \frac{2}{(\theta-1)^3} = \frac{2}{(\theta-1)^2}$$

$$I(\theta) = I_1 + I_2 + I_3 = \frac{1}{(\theta-1)^2} \quad \text{— кепс на } (1, \infty) \quad \text{и } I(\theta) > 0$$

моу регулярна

Для сильной регулярности осталось:

$$5) \frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{\infty} dF(x, \theta) = \int_{-\infty}^{\infty} d \frac{\partial^2 F(x, \theta)}{\partial \theta^2}$$

$$\frac{\partial^2}{\partial \theta^2} \int_1^{\infty} p(x, \theta) dx = 0.$$

$$\int_1^{\infty} \frac{\partial^2 p(x, \theta)}{\partial \theta^2} dx = \int_1^{\infty} \left(-\frac{1 \ln x}{x^\theta} + \frac{\ln^2 x (\theta-1)}{x^\theta} \right) dx = \dots =$$

$$= \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = 0.$$

=) сильн. регулярна

Тогда мет. макс. правд.: $\text{сост.}, \text{ас. несмещ.}, \text{ас. норм.}, \text{ас. эфф.}$
 $F(x) = \int_1^x \frac{\theta-1}{t^\theta} dt = \frac{1}{x^{\theta-1}} \cdot 1 = \frac{1}{2}$
 $p = 0.5, F(x_{0.5}) = 0.5$

$$X_{(\theta)} = 2 \frac{1}{\theta-1}$$

$$g(\theta) = 2 \frac{1}{\theta-1}$$

$$v(g(\theta)) = 2 \frac{1}{\theta-1} \cdot \ln 2 \cdot -\left(\frac{1}{(\theta-1)^2}\right)$$

$$\sigma^2(\theta) = v^T g(\theta) \cdot J^{-1}(\theta) \cdot v g(\theta) = \left(2 \frac{1}{\theta-1} \ln 2 \cdot -\frac{1}{(\theta-1)^2}\right)^2 \cdot (\theta-1)^2$$

тогда

$$\sqrt{n} \left(\frac{2 \frac{1}{\bar{\theta}-1} - 2 \frac{1}{\theta-1}}{2 \frac{1}{\bar{\theta}-1} \cdot \ln 2 \cdot \frac{1}{\bar{\theta}-1}} \right) \sim N(0,1)$$

$$t_1 = U_{\frac{1-\beta}{2}}, \quad t_2 = U_{\frac{\beta}{2}}$$

$$P\left(t_1 < \frac{\sqrt{n}(\bar{\theta}-1) \cdot (2 \frac{1}{\bar{\theta}-1} - 2 \frac{1}{\theta-1})}{2 \frac{1}{\bar{\theta}-1} \cdot \ln 2} < t_2\right) = \beta$$

$$\Rightarrow \left[2 \frac{1}{\bar{\theta}-1} - \frac{2 \frac{1}{\bar{\theta}-1} \ln 2}{\sqrt{n}(\bar{\theta}-1)} t_2 < X_{(\theta)} < 2 \frac{1}{\bar{\theta}-1} - \frac{2 \frac{1}{\bar{\theta}-1} \ln 2}{\sqrt{n}(\bar{\theta}-1)} t_1 \right]$$

асимп. дов. интер. для θ

асимп. интервал для θ

$$\sqrt{n} \frac{\bar{\theta} - \theta}{\sigma(\theta)} \sim N(0,1)$$

$$g(\theta) = \theta, \quad v g(\theta) = 1.$$

$$\sigma^2(\theta) = v^T g(\theta) \cdot J^{-1}(\theta) \cdot v g(\theta) = 1 \cdot (\theta-1)^2$$

По лемме Смулского (θ мен на $\bar{\theta}$)

$$\frac{\sqrt{n}(\bar{\theta} - \theta)}{\bar{\theta} - 1} \sim N(0,1)$$

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$$t_1 \frac{\bar{\theta} - 1}{\sqrt{n}} < \bar{\theta} - \theta < t_2 \frac{\bar{\theta} - 1}{\sqrt{n}}$$

$$\left[\bar{\theta} - \frac{\bar{\theta} - 1}{\sqrt{n}} t_2 < \theta < \bar{\theta} - \frac{\bar{\theta} - 1}{\sqrt{n}} t_1 \right]$$